

Assessment Tool 1 – Answers

Experiment 1: Decision Tree Classification (Iris Dataset)

(a) Load, Pre-process, Split

Dataset: Iris (150 rows, 4 numeric features, 3 classes). No missing values found.

First 5 rows:

sepal length	sepal width	petal length	petal width	species
5.1	3.5	1.4	0.2	setosa
4.9	3.0	1.4	0.2	setosa
4.7	3.2	1.3	0.2	setosa
4.6	3.1	1.5	0.2	setosa
5.0	3.6	1.4	0.2	setosa

Data types: all four feature columns are float64; target/species is categorical (encoded 0/1/2). No missing values (0 nulls in every column), so no imputation was needed. Categorical encoding: species label-encoded to target (0=setosa, 1=versicolor, 2=virginica).

Split: 80% train / 20% test, stratified on class (train=120, test=30), random_state=42.

(b) Build Decision Tree (Gini)

Tree structure (top splits):

petal length $\leq$ 2.45 -> class: setosa

petal length $>$ 2.45

petal width $\leq$ 1.65

petal length $\leq$ 4.95 -> versicolor

petal length $>$ 4.95 -> mostly virginica (splits further on sepal length/width)

petal width $>$ 1.65

petal length $\leq$ 4.85 -> splits on sepal width (mostly virginica)

petal length $>$ 4.85 -> virginica

Root node: petal length (threshold 2.45 cm). Justification: Gini feature-importance scores were petal length = 0.559, petal width = 0.406, sepal width = 0.029, sepal length = 0.006. Petal length gives the purest split (perfectly separates setosa from the other two species with zero impurity), so it produces the largest Gini gain and is chosen as the root.

(c) Evaluation

- **Accuracy: 93.3%** (28/30 correct on the test set)
- **Confusion matrix:**

	Pred setosa	Pred versicolor	Pred virginica
True setosa	10	0	0
True versicolor	0	9	1
True virginica	0	1	9

- **Classification report:** setosa precision/recall/F1 = 1.00/1.00/1.00; versicolor = 0.90/0.90/0.90; virginica = 0.90/0.90/0.90; macro/weighted avg = 0.93.

Comment: Setosa is perfectly separated because it is linearly distinguishable on petal length alone. All errors occur between versicolor and virginica, which overlap in petal dimensions.

Misclassified instances: 1. Row (sepal 6.1, sepal-width 2.6, petal 5.6, petal-w 1.4) — true label **virginica**, predicted **versicolor** (petal length/width fall near the boundary region). 2. Row (sepal 6.7, sepal-width 3.0, petal 5.0, petal-w 1.7) — true label **versicolor**, predicted **virginica** (petal width 1.7 is just above the 1.65 split threshold, pushing it into the virginica branch).

Experiment 2: Reinforcement Learning – Q-Learning (4×4 Grid World)

(a) Environment Definition

- **States:** 16 cells, numbered 0–15 (row-major, 4×4 grid).
- **Actions:** Up, Down, Left, Right (moves that hit a wall keep the agent in place).
- **Rewards:** +10 at goal (state 15), -5 at obstacle (state 10), -1 for every other step.
- **Q-table:** initialized to all zeros, shape (16 states × 4 actions).
- **Hyperparameters:** learning rate $\alpha = 0.1$, discount factor $\gamma = 0.9$, exploration rate $\epsilon = 0.2$ (ϵ -greedy).

(b) Q-Table Evolution (100 episodes)

Q-values for two representative states across training:

Episode	State 0 [Up, Down, Left, Right]	State 9 [Up, Down, Left, Right]
1	[-0.20, -0.37, -0.20, -0.10]	[-0.20, -0.19, -0.22, -0.50]
50	[-2.05, -2.03, -2.12, -2.04]	[-0.55, 3.31, -0.65, -1.36]
100	[-1.89, -2.06, -2.12, 0.79]	[-0.34, 6.07, -0.18, -1.85]

The Q-table starts near zero (episode 1), develops a clear preference for the goal-directed action by episode 50 (e.g., state 9 strongly favors “Down”), and by episode 100 the values have grown larger and more stable — the “Right” action at state 0 and “Down” at state 9 clearly dominate, showing the policy has converged toward the shortest safe path to the goal.

(c) Final Policy and Convergence

Learned optimal action per state (0 = start, 15 = goal, 10 = obstacle):

0:Right 1:Down 2:Down 3:Down
4:Down 5:Down 6:Right 7:Down
8:Right 9:Down 10:OBST 11:Down
12:Right 13:Right 14:Right 15:GOAL

The policy routes the agent around the obstacle at state 10 and down/right toward the goal at state 15.

Cumulative reward per episode: average reward for the first 10 episodes ≈ -18.3 (mostly random exploration, frequent wall bumps and long paths); average reward for the last 10 episodes $\approx +1.9$ (short, mostly direct paths to the goal, occasionally still exploring due to $\epsilon=0.2$).

Convergence justification: the steadily rising per-episode reward and the stabilizing Q-values (episode 50 $\rightarrow$ 100 changes shrink) show the Bellman updates are converging toward the optimal action-value function; residual variance is expected because ϵ -greedy still explores 20% of the time.